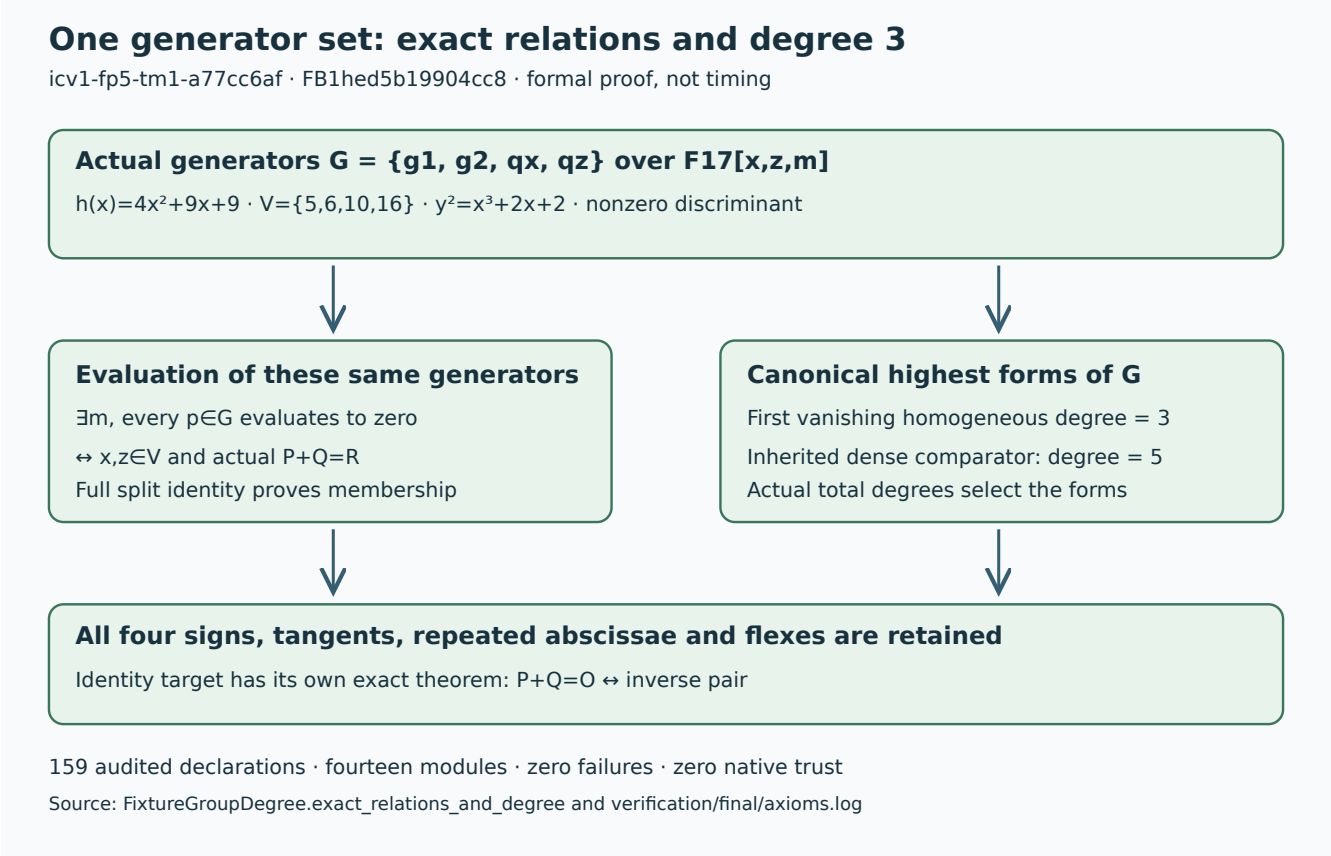
Concrete F17 relations and canonical degree

2026-10-31 · checked formalization continuation

The theorem `FixtureGroupDegree.exact_relations_and_degree` proves the actual four-generator system equivalent to signed factor-base relations in Mathlib's elliptic-curve group. The same generator set has canonical homogeneous first-vanishing degree 3, while the inherited dense comparator has degree 5. Each valid finite target and all four signs are included.



Curve `icv1-fp5-tm1-a77cc6af`, base `FB1hed5b19904cc8`. Over F17, the curve is $y^2=x^3+2x+2$, the graph is $h(x)=4x^2+9x+9$ and $V=\{5,6,10,16\}$. These existing identities and the FB1 object are reused; the fixture's existing unregistered status is unchanged.

The split identity is proved as a polynomial equality, rather than inferred from finite-field function values. Every generator evaluation is proved equal to its named secant residual. Tangents, repeated abscissae and flexes remain covered. The identity target is separately exact: inverse pairs.

Proof scope and verification

The headline theorem joins three checked statements:

1. For every x,z, some slope annihilates all original generators exactly when the signed base points have actual sum R.
2. The canonical highest-form ideal of those same generators first vanishes in homogeneous degree 3.
3. The inherited dense comparator first vanishes in homogeneous degree 5.

The actual total degree of each polynomial selects its highest form. Nonzero discriminant and the concrete complete-splitting identity are proved in Lean. Every target on the concrete curve is nonsingular. All ordinate lifts are represented by the four sign branches.

Checks

159 declarations in fourteen modules pass the unchanged transitive-axiom auditor: zero failures and zero native trust. The new module passes warnings-as-errors. The prior published three-module group bridge also rebuilds and passes its 22-declaration audit and warnings-as-errors, closing the earlier outage gap. The original recovered source matches the published copy byte-for-byte.

Lean 4.19.0; Mathlib revision `c44e0c8ee63ca166450922a373c7409c5d26b00b`. Nine Git dependencies match the lockfile; thirty inputs are hashed. The two sealed prior proof projects are unchanged relative dependencies. All axioms are among propext, Classical.choice and Quot.sound. lean4checker was unavailable. Seven failed development builds and all successful receipts remain in the evidence archive.

Interpretation limits

The degree is the first vanishing homogeneous quotient degree. No affine F4 degree, natural yield or runtime is measured here. This continuation does not add an independent dense S3-to-group equivalence theorem or claim external novelty. Existing quantitative figures and leaderboard rows stay unchanged. The figure documents formal coverage, not measured speed.

Sources and replay

The exact declaration/hypothesis map is ../FORMALIZATION.md. Run `bash verify.sh /tmp/secant-fixture-receipts` from the project root. See ../verification/final/axioms.log, ../verification/metadata.json and the source hashes for the complete records.

The concrete fixture is ../../secant/fixture.json. The copied identity certificate is IDC1h621ed1a51b5f3ca8; its evaluated-tier status is retained. The actual group reuses Mathlib and Angdinata–Xu, ITP 2023: https://doi.org/10.4230/LIPIcs.ITP.2023.6.